From information coefficient to Sharpe ratio: pre-registered tests of machine-learning and graph-network return predictors on a free US equity cross-section

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Abstract

We ask whether a cross-sectional return predictor fitted to free, point-in-time US equity data becomes a tradable long-short book, and whether message passing over firm graphs adds information to it. The protocol is pre-registered: a fixed shallow boosted-tree regressor on 28 characteristics, yearly expanding refits with an embargo, a development window 2009 to 2018, a check window 2019 to 2022, one hold-out read of 2023 and 349 pre-declared portfolio cells. For the graph question, relational GraphSAGE and graph-attention networks are trained end to end on SIC, customer-supplier, input-output and return-correlation graphs and compared by paired ΔIC with an identical no-graph network and, as monthly and quarterly IC levels only, with the same network on degree-preserving rewired graphs. We find, first, that the rank signal is real and small and that it replicates out of window: weekly rank IC $+0.024$ on development, $+0.051$ on the check window, monthly $+0.087$ on the 2023 hold-out. Second, no cell reaches a net Sharpe ratio of 1.5 on both windows, and since no cell's gross Sharpe ratio reaches 1.5 on either window, no cost level can close the gap. The selected cell nets 0.63 ± 0.32 on development (1.9 % a year, maximum drawdown -11.9 %) and 0.28 ± 0.51 on the check window (1.7 %, -8.8 %), and the least demanding cell would need at least about $2.50\times$ its information. Third, on the development window the three completed graph networks add a paired ΔIC that cannot be told from zero on any horizon. Because confirming a Sharpe-0.3 property at its own noise level would take decades, how much of such a signal to hold has to come from a prior rather than from the data.

1 Introduction

A model that ranks stocks by next-period return is judged by two numbers. The first is its information coefficient (IC), the cross-sectional correlation between its scores and realised returns, which is well documented and mostly encouraging (Gu et al., 2020; Chen et al., 2024; Kelly et al., 2024). The second is the net Sharpe ratio of a book built from the scores, which is less well documented and less encouraging (Novy-Marx and Velikov, 2016; Avramov et al., 2023). The fundamental law of active management links them (Grinold, 1989; Clarke et al., 2002): a Sharpe ratio is approximately the IC times the square root of breadth, the number of independent bets, times a transfer coefficient set by the portfolio construction, less a cost drag set by turnover. We measure each term of that identity on one dataset under a pre-registered protocol, which lets us answer a practical question: can a model built from free data reach a net Sharpe ratio of 1.5 with drawdown under 25%, and do graph methods help?

Every input is free, with prices from a public survivor-only feed, fundamentals from SEC XBRL filings, earnings dates from 8-K filings, industry graphs from SIC codes and the BEA input-output tables, and customer-supplier links from filing text. A student can reproduce all of it, but the free price feed brings a survivorship bias that we bound in Section 10 and cannot remove.

Contributions.

1. A shallow booster with hyperparameters fixed in advance carries a rank signal whose sign replicates out of its development window and on one held-out year, whereas selecting hyperparameters per refit on a noisy validation slice removes its advantage over a linear control.

2. Along a 55-cell grid of constructions, an exact cost-ladder identity shows that the shortfall against the target comes from too little information rather than from cost. No gross Sharpe ratio reaches 1.5, so no cost level closes the gap, and to first order the least demanding cell would need at least about 2.50× its IC.
3. On the development window, the three completed relational graph networks, trained end to end on point-in-time firm graphs, add a paired ΔIC against an identical no-graph network that cannot be told from zero, and the rewired-graph comparison, which we completed only as monthly and quarterly IC levels, points the same way (Maslov and Sneppen, 2002).
4. A power calculation puts confirmation of a Sharpe-0.3 property decades away, which leaves the sizing decision to a prior.

Related work. Trees and shallow networks are the strongest cross-sectional predictors on CRSP, with small out-of-sample R^2 and large pre-cost decile spreads (Gu et al., 2020), much of which costs and microcaps remove (Avramov et al., 2023). Economic links predict firm returns (Cohen and Frazzini, 2008; Menzly and Ozbas, 2010), and graph neural networks (Kipf and Welling, 2017; Hamilton et al., 2017; Veličković et al., 2018; Schlichtkrull et al., 2018) are their end-to-end form. We know of few tests of them against a rewired-graph null on a point-in-time panel, and ours is partial (Section 10). For inference under selection we follow Harvey et al. (2016), Bailey and López de Prado (2014) and Hansen (2005), for Sharpe-ratio uncertainty Lo (2002) and Ledoit and Wolf (2008), and we read a small signal against the documented decay of published anomalies (McLean and Pontiff, 2016; Hou et al., 2020).

Figure 1 (p. 5) gives the pipeline in one picture. Sections 2 to 7 define the panel, validation geometry, predictors, graphs, book construction and statistics; Sections 8 to 10 report results, power and limitations.

2 Data and the point-in-time panel

Universe and formation grid. Formation dates t are weekly (W, every fifth session from 2004-01-02), monthly (M, last session) or quarterly (Q, last session of each quarter). "ALL" is every US common stock with a bar on t, raw close $\geq$ \$5 and 63-session median dollar volume $\geq$ \$1M. "LIQ" is the half of ALL above the within-date median of that volume, and "TOP500" the 500 largest by it. Prices are survivor-only because the feed lists firms as of October 2026 (Section 10). $r_{i,s}$ is the daily adjusted-close return (missing if above 3 or below -0.95) and M_s the equal-weight mean return of ALL fixed at the latest month-end before s. The panels run to 2022-12 with 1,664,463 weekly, 397,417 monthly and 132,961 quarterly firm-dates, of which 1,347,743 / 318,475 / 103,945 receive out-of-sample predictions.

Characteristics. Table 1 lists the 28 characteristics at formation row t. Fundamentals use SEC facts accepted at or before the close of t, go stale after 18 months, and are set missing before 2009-01-01 and on rows with implied daily turnover above 10%; trailing twelve months (TTM) is the sum of four quarters or the latest annual. Earnings events use 8-K item 2.02 filings with a 16:00 ET session rule.

Rank transform. Per date and characteristic, finite values are winsorised at the 1st and 99th percentiles, ranked with ties averaged ($R_i \in \{1, \dots, n\}$) and mapped to

$$\tilde{x}_{i,f,t} = \frac{R_i - 1}{n - 1} - \frac{1}{2} \in [-\tfrac{1}{2}, \tfrac{1}{2}], \qquad m_{i,f,t} = 0; \qquad \text{missing: } \tilde{x} = 0,\ m = 1. \tag{1}$$

The input is $x_{i,t} = (\tilde{x}_{i,1..28},\ 11 \text{ SIC one-hots},\ m_{i,1..28}) \in \mathbb{R}^{67}$; the one-hots (ten divisions plus "unknown") come from the current filer profile and are static. A 29th candidate (gross profitability) was dropped by a 30% coverage rule fixed before any result was read.

Target. With u the next formation row, entry at the close of row t+1 and exit at the close of row u+1,

$$y^{raw}_{i,t} = \prod_{s=t+2}^{u+1} (1 + r_{i,s}) - 1, \qquad y^{rank}_{i,t} = \frac{R(y^{raw}_{i,t}) - 1}{n_t - 1} - \frac{1}{2}, \qquad y^{ex}_{i,t} = y^{raw}_{i,t} - \overline{y^{raw}_t}, \tag{2}$$

ranks over ALL with ties averaged. Models fit y^{rank}, ICs are scored against y^{ex} and books are paid in y^{raw}. A within-date rank is not market-neutral, since with $r_{i,t} = a_t + (\beta_i - \bar{\beta})\,\mathrm{mkt}_t + \varepsilon_{i,t}$ it keeps the

Table 1: The 28 characteristics at formation row t. Lookbacks are trading sessions; "min k" is the minimum number of non-missing observations. DV = close × volume.

Name	Definition
`mom_12_1`, `mom_6_1`	compounded return close($t-252$) →close($t-21$); close($t-126$) →close($t-21$)
`rev_1m`, `rev_1w`	close($t-21$) →close(t); close($t-5$) →close(t)
`vol_63`	sample s.d. of r over 63 sessions (min 50)
`ivol_63`	$\sqrt{\max(0, \mathrm{Var}(r) - \mathrm{Cov}(r, M)^2/\mathrm{Var}(M))}$, 63 sessions (min 50 pairs)
`maxret_21`	max r over 21 sessions (min 15)
`amihud_63`	mean of $\lvert r_s\rvert/(DV_s/10^6)$ over 63 sessions (min 50)
`log_dv_63`	log of the 63-session median DV (min 50)
`beta_252`	$\mathrm{Cov}(r, M)/\mathrm{Var}(M)$ over 252 sessions (min 200 pairs)
`skew_63`	$m_3/m_2^{3/2}$ over 63 sessions (min 50)
`hi_252`	adjusted close$_t$ / 252-session maximum (min 200)
`log_price`	log of the raw traded close
`days_since_ea`	calendar days since the latest 8-K item 2.02 day $0 \le t$ (missing if > 400)
`days_to_exp_ea`	days until (day 0 of the first event within the last 364 days) +364
`ann_ret_last`	$\sum_{k=-1}^{1}(r_{d+k} - m_{d+k})$ around the latest clean event day d, m = total market return
`ea_timing`	+1 / 0 / −1: latest expected announcement late / on time / early (5-session tolerance)
`log_mcap`	$\log(P_t^{raw} S_t)$, S_t = cover-page share count on t's split basis
`bm`, `ep`, `sp`	book equity, TTM net income, TTM revenue, each over market capitalisation
`asset_growth`	assets / assets twelve months earlier −1
`accruals`	(TTM net income − TTM operating cash flow) / assets
`nsi`	$\log(S_t/S_{old})$, S_{old} = share count known twelve months earlier
`leverage`, `cash_assets`	liabilities / assets; cash / assets
`roe`	TTM net income / book equity
`sue`	$(\mathrm{EPS}_q - \mathrm{EPS}_{q-4})$/s.d. of the eight previous seasonal differences (need ≥ 6)

term $(\beta_i - \bar{\beta})\,\mathrm{mkt}_t$, and the monthly rank target has correlation 0.75 with the market excess return; Section 6 removes this at the portfolio stage.

Zones. The data are partitioned once into a training zone (2004-01 to 2023-01), a walk-forward zone (2023-01 to 2024-07) and a sealed zone from 2024-07-01 from which no price or return has been read. Inside the training zone, 2009 to 2018 is the development window on which every selection was made (some later design choices were proposed after earlier check-window results had been reported, which Section 3 counts as reuse) and 2019 to 2022 the check window, which this programme had read three times before the graph experiments (counted in Section 3). The walk-forward zone was read once, for 2023, under a rule frozen in advance.

3 Validation geometry

Expanding refits with an embargo. Let $mi = 12 \cdot \mathrm{year} + \mathrm{month} - 1$ index the *entry* month of a label. For each out-of-sample year $Y \in \{2009, \ldots, 2022\}$ a new model is fitted on data strictly before Y, with embargo $e = 1$ month and a 12-month validation slice:

$$\mathrm{test} = [12Y,\ 12Y{+}11], \qquad \omega = 12Y - 1 - e, \qquad \mathrm{val} = [\omega - 11,\ \omega], \qquad \mathrm{fit} = [m_0,\ \omega - 13], \tag{3}$$

m_0 = February 2004, with a one-month gap between fit and validation. The condition that has to hold is one on daily returns rather than on month labels. With entry(l) the close at which a label's holding window starts (row $t{+}1$) and exit(l) the close at which it ends (row $u{+}1$),

$$\max_{l \in \mathrm{fit} \cup \mathrm{val}} \mathrm{exit}(l) \ \le\ \min_{l \in \mathrm{test}} \mathrm{entry}(l), \qquad \max_{l \in \mathrm{fit}} \mathrm{exit}(l) \ \le\ \min_{l \in \mathrm{val}} \mathrm{entry}(l), \tag{4}$$

so no two labels on opposite sides of a boundary share a daily return. The same $e = 1$ and validation slice are used at every horizon; eq. (4) then holds by arithmetic on entry and exit rows (strictly at the weekly and monthly horizons, with equality of exit and entry closes at the quarterly one). This is a property of the construction, and only the earlier window-grid study's splitter asserts it in code. Refits retrain from scratch with seed = base seed + first test month index.

Graph-network masks. The graph experiments use no validation slice: test rows have $12Y \le mi \le 12Y{+}11$; training rows have $mi \le 12Y - 2$ and exit_date ≤ the earliest test formation date. They train through November of $Y{-}1$ (the booster's fit slice ends in October of $Y{-}2$); both respect eq. (4).

What "no selection" means. The headline booster's depth, learning rate, rounds and subsampling were fixed before the out-of-window reads, but the configuration itself was chosen as the best of three model classes on development, with a development IC of +0.031 on the original panel as the best of five seeds (+0.029 averaged) and +0.023 on the corrected panel of Table 3 (cause not isolated). Fixing the hyperparameters matters, because with per-refit selection over a 144-cell grid on the 12-month validation IC the same class reads +0.014, about ridge's +0.013, since the validation IC has a standard error of about 0.024 while the selected grid point's margin over the next best is 0.000 to 0.005. The graph networks' single hyperparameter set was hashed and registered before any graph result was read; the code refuses unregistered hashes.

Trial counting. The portfolio programme has $N = 349$ pre-declared cells: 294 in a first programme (234 monthly cells with a 2019 to 2022 check window and 60 weekly cells on panels ending 2018) and a 55-cell grid on panels rebuilt through 2022, first run on development only on the older panels and counted once. Each graph-network prediction set evaluated adds 55, so the graph programme counts $N = 349 + 55k$. Section 7 turns N into an expected maximum of null Sharpe ratios and a deflated Sharpe ratio (Bailey and López de Prado, 2014). The check window had been read three times before the graph experiments (the out-of-window replication, an earlier graph-feature study, the 349-cell programme), so every pass on it is a pass on a reused window.

4 Predictors

All models map $x_{i,t} \in \mathbb{R}^{67}$ to a score $\hat{y}_{i,t}$ and are fitted to y^{rank}. Where a library's convention differs from the textbook form, the computed form is given.

Ridge. $\hat{\beta} = \arg\min_\beta \|y - b\mathbf{1} - X\beta\|_2^2 + \alpha\|\beta\|_2^2$ with an unpenalised intercept, α chosen on the validation slice; two controls, three columns (`mom_12_1`, `sue`, `log_mcap`) and all 67.

Gradient-boosted trees (fixed booster). XGBoost (Chen and Guestrin, 2016) fits $\hat{y}^{(k)} = \hat{y}^{(k-1)} + \eta f_k$ by minimising, with g_i, h_i the first and second derivatives of the loss at $\hat{y}_i^{(k-1)}$,

$$\mathcal{L}^{(k)} \approx \sum_i \Big[g_i f_k(x_i) + \tfrac{1}{2} h_i f_k(x_i)^2 \Big] + \gamma T + \tfrac{1}{2}\lambda \sum_{j=1}^{T} w_j^2, \qquad w_j^\star = -\frac{G_j}{H_j + \lambda}, \tag{5}$$

G_j, H_j the sums over leaf j; a split needs $H \geq$ `min_child_weight` in both children, which under squared error ($h_i = 1$) means a row count. The fixed booster has depth 3, $\eta = 0.05$, 300 rounds, row and column subsampling 0.8, `min_child_weight` 100 and 64-bin histograms, uses the library defaults for λ, γ, and has no grid and no early stopping. The earlier, selected version chose among eight settings with up to 2000 rounds and early stopping on validation IC.

Pairwise rank objective. This variant fits the same trees to the pairwise logistic loss $\sum_{y_i > y_j} \log(1 + e^{-(\hat{y}_i - \hat{y}_j)})$ within each date (Burges et al., 2005), with eight sampled pairs per row. Its hessians are small, so a `min_child_weight` copied from the squared-error grid forbids every split and returns a constant (Appendix A); the fixed version uses 1.

Feed-forward network (NN3). Following Gu et al. (2020), with $h^0 = x$,

$$h^l = \mathrm{ReLU}\big(\mathrm{BN}(W_l h^{l-1} + b_l)\big),\ l = 1, 2, 3 \text{ (widths } 32, 16, 8), \qquad \hat{y} = w_4^\top h^3 + b_4, \tag{6}$$

trained on $\frac{1}{B}\sum(\hat{y}_i - y_i)^2 + \lambda \sum_l \|W_l\|_1$ with Adam, batches of 10^4, at most 100 epochs, patience 5 on validation IC; the chosen (λ, lr) is completed to ten seeds and averaged.

Cross-sectional transformer. One date is one set: $X_t \in \mathbb{R}^{N_t \times 67}$ is the sequence, with no positional encoding, so the map is permutation-equivariant over firms (Vaswani et al., 2017). $H^0 = X_t W_{in} + b$ with $d_{model} = 64$; two pre-norm layers, four heads, feed-forward width 128,

$$H \leftarrow H + \mathrm{MHA}(\mathrm{LN}(H)), \qquad H \leftarrow H + \mathrm{FFN}(\mathrm{LN}(H)), \qquad \mathrm{head}_a = \mathrm{softmax}\Big(\frac{Q_a K_a^\top}{\sqrt{16}}\Big) V_a, \tag{7}$$

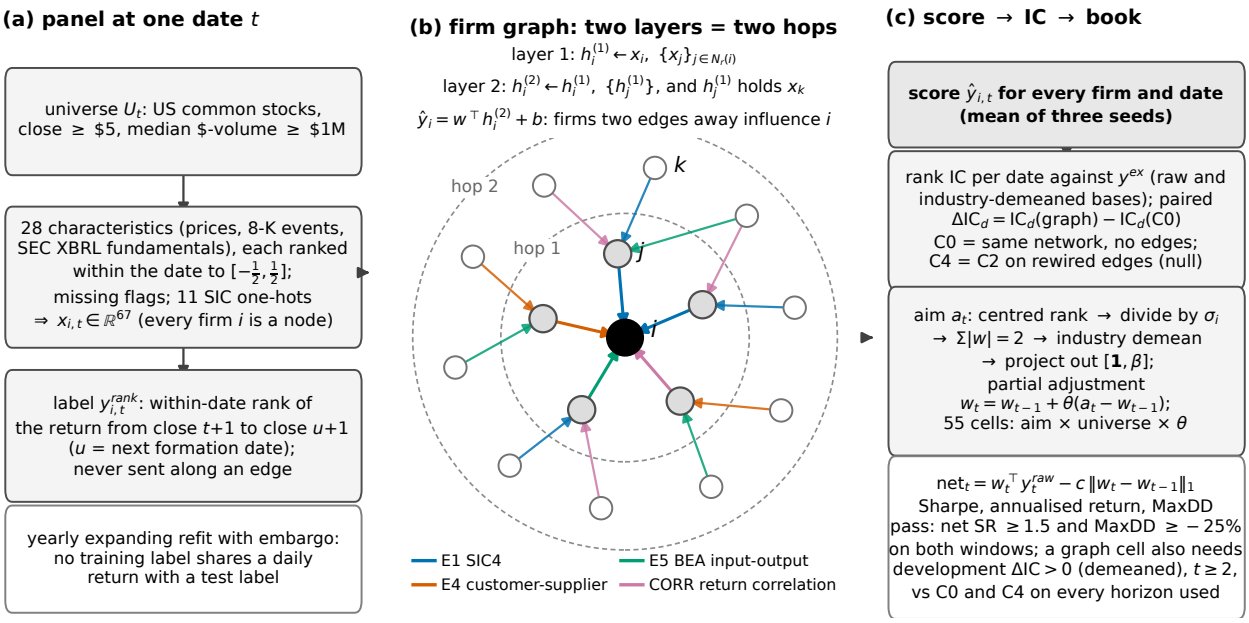


Figure 1: The pipeline. (a) Each formation date is one cross-section: the ranked characteristics of eq. (1) are the node input, the label of eq. (2) is the within-date rank of the next-period return and never travels along an edge. (b) The four relations of Table 2 define directed edges $j \to i$. Two layers of eq. (8) (or eq. (9)) give a two-hop receptive field: layer 1 aggregates the features of the in-neighbours $N_r(i)$, layer 2 aggregates their layer-1 states, so a firm k two edges away reaches $\hat{y}_i$ through $h_j^{(1)}$. (c) Scores are compared by paired ΔIC with the identical network without edges (C0) and with the four-relation network C2 trained on rewired edges (C4), and become a book through eqs. (12) to (15).

and $\hat{y}_t = \text{Linear}(\text{LN}(H^{(2)}))$, so each firm's score depends on every firm of its date. It is trained on MSE per date with Adam at 10^{-3}, at most 3000 firms per date, at most 20 epochs and patience 3, and the score is the mean of five seeds.

Rank-mean ensemble and composites. The ensemble averages the within-date ranks of eq. (1) of its components. Composites $c_{i,t} = \frac{1}{|F|}\sum_{f \in F} s_f \tilde{x}_{i,f,t}$ with no fitted magnitudes serve as benchmarks: (a) 23 characteristics with literature signs fixed before any number was read, (b) the textbook five (momentum +, book-to-market +, size −, profitability +, issuance −), (c) signs re-estimated each year from months up to $12Y - 2$.

Graph networks. Node input is $x_i \in \mathbb{R}^{67}$ with the 28 rank columns scaled by $\sqrt{12}$ (unit variance for a uniform on $[-\frac{1}{2},\frac{1}{2}]$). The networks have two layers of hidden width 64 and dropout 0.1, and the attention cell uses four heads. A relation r gives, per date, directed edges $j \to i$ with weight w_{ij}^r over the firms present; $N_r(i)$ is the set of sources into i, and the seven cells below differ in the relations they use and in how messages are aggregated.

C0, MLP (baseline).
$h^{(l)} = \text{Drop}(\text{ReLU}(W_l h^{(l-1)} + b_l))$, $l = 1, 2$; $\hat{y}_i = w^\top h^{(2)} + b$. It uses the same features, training loop, epochs and seeds as every graph cell.

C0g, aggregates.
The same MLP on $[x_i,\ \{\bar{x}_i^r\}_{r=1}^4,\ \{\log(1 + \deg_r(i))\}_{r=1}^4] \in \mathbb{R}^{339}$, $\bar{x}_i^r$ the weighted neighbour mean of eq. (8), so graph information enters as fixed features.

C1, C2, relational GraphSAGE-mean
(Hamilton et al., 2017; Schlichtkrull et al., 2018):

$$h_i^{(l+1)} = \text{Drop}\Big(\text{ReLU}\Big(W_s^{(l)} h_i^{(l)} + b^{(l)} + \sum_r W_r^{(l)} \frac{1}{\max(1, |N_r(i)|)} \sum_{j \in N_r(i)} w_{ij}^r\, h_j^{(l)}\Big)\Big), \tag{8}$$

$\hat{y}_i = w^\top h_i^{(2)} + b$; weights enter the message, the divisor is the unweighted in-degree, a node without in-edges contributes zero. C1 uses E1, E4, E5; C2 adds CORR.

Table 2: The four relations. Edges are directed $j \to i$ with weight w_{ij}; the signed relations store both directions and encode direction in the sign.

Relation	Edge rule (as built)	Weight w_{ij}
E1 SIC4	two firms whose filers share a four-digit SIC code on that date; in groups of more than 21 firms each firm samples 20 others	1
E4 customer-supplier	(customer, supplier) pairs resolved from filing text with a point-in-time name resolver (precision 0.96 on 200 hand-checked edges), valid between filing dates	$+1$ customer→supplier, -1 reverse
E5 BEA input-output	firm SIC4 → BEA industry; industry pairs are the top-10 buyers of each seller and top-10 suppliers of each buyer in the BEA use table of year year$(d) - 2$; one direction per pair (the larger share); no within-industry edges; each firm samples at most 20 firms of neighbouring industries with probability proportional to share	$\pm$ share, supplier→customer positive
CORR	252 sessions ending at d, returns minus the daily cross-sectional mean, pairwise correlation with at least 189 overlapping sessions; the $k = 10$ most positively correlated firms point into each firm	correlation

C3, relational graph attention

(Veličković et al., 2018): for relation r, $g_j = W_r h_j \in \mathbb{R}^{64}$ is split into four head blocks $g_j^{(q)} \in \mathbb{R}^{16}$ with learned $\mathbf{a}_{r,q}^{src}, \mathbf{a}_{r,q}^{dst} \in \mathbb{R}^{16}$,

$$e_{ij}^{(q)} = \mathrm{LeakyReLU}_{0.2}(\mathbf{a}_{r,q}^{src\top} g_j^{(q)} + \mathbf{a}_{r,q}^{dst\top} g_i^{(q)}), \qquad \alpha_{ij}^{(q)} = \frac{\exp e_{ij}^{(q)}}{\sum_{k \in N_r(i)} \exp e_{ik}^{(q)}},$$
$$z_i = W_s h_i + b + \sum_r \Big\|_{q=1}^{4} \sum_{j \in N_r(i)} \alpha_{ij}^{(q)} g_j^{(q)}, \tag{9}$$

$h_i' = \mathrm{Drop}(\mathrm{ReLU}(z_i))$. The attention ignores w_{ij}, so the signed relations become undirected in C3.

C4, rewired null.

C2 trained on the rewired edge tables of Section 5.

C5, temporal.

Per firm a GRU (Cho et al., 2014) (input $68 = 67$ features plus a presence mask, hidden 64) over the last twelve snapshots; its final state is the node input of eq. (8).

For each refit year Y and seed $s \in \{0, 1, 2\}$ a fresh model (seed $1000s + Y$) minimises, one date per Adam step,

$$L_t = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} (\hat{y}_i - y_{i,t}^{rank})^2, \qquad \text{Adam, lr } 10^{-3}, \text{ weight decay } 10^{-5}, \text{ 15 epochs}, \tag{10}$$

over the training firms $\mathcal{S}_t$ with a finite target on date t, dates shuffled each epoch, no early stopping, no validation-based choice. All ICs and books use the mean of the three seeds.

5 Graphs and their nulls

Relations. Four point-in-time relations are edge tables with validity intervals; an edge is used on date d iff valid_from $\le d <$ valid_to (Table 2). Only features and hidden states pass along edges; no neighbour's target is ever aggregated.

Rewired-graph null. Whether the topology carries information beyond degree and group membership is addressed by training the same network on graphs that preserve those quantities and destroy the pairing (Maslov and Sneppen, 2002), although the paired test against it was not completed (Section 8.4). One rewired table per relation is drawn per seed and held fixed through time. For E1, SIC4 timelines are permuted among firms with the identical SIC2 timeline (SIC2 kept exactly, SIC4 group sizes approximately, and about 12% of firms keep their label). For E4, degree-preserving double-edge swaps act on the unique (customer, supplier) pairs within customer-SIC4 strata (fallback SIC3, SIC2), with realised swap shares of 0.30 to 0.39 per snapshot. For E5, the use table is permuted within columns per data year, so each buying industry keeps its supplier count. For CORR, each firm and year gets a fixed random ranking of the other firms by stratum (SIC2 × beta tercile, then SIC2, division, any),

and on each date the in-neighbours are the best-ranked present firms, as many as the real in-degree, with the real sorted weights reassigned.

Label-shift diagnostic. A model that has memorised firm identity, or a slowly varying attribute leaking through the features, predicts last year's label about as well as this year's. For each refit every training row (firm f, formation i) is relabelled with f's label at formation $i-K$ (else $i-K-1$, else $i-K+1$), $K = 54/13/5$ for W/M/Q, valid only if that source window ended at least 366 days before the row's formation; rows without a valid source are dropped. The declared read rule is that the signal "persists under shift" if the shifted development IC is positive, has $t \geq 2$ and is at least half the real IC, and a pass that persists would be relabelled "not attributable to current information".

6 From predictions to a book

Decile book. Bin each date by the percentile rank of the prediction, hold the top decile long and the bottom decile short with equal weights within a leg (one dollar each side), and pay cost c per side on the two-sided turnover $\tau_t = \|w^{new} - w^{prev}\|_1$ against the previous target weights:

$$g_t = \bar{y}_{long} - \bar{y}_{short}, \qquad \text{net}_t = g_t - c\,\tau_t. \tag{11}$$

This is the book of Section 9 and Table 6; the earlier window-grid study's decile book drifted the previous weights by the holding return, $w_i^{drift} = w_i(1+y_i)/(1+\bar{y}_{leg})$, before differencing.

The grid construction. The grid construction scales each name by its inverse volatility and controls the book's industry and market-beta exposure. Let m_t be the tradable set (present, finite score, universe filter, at least 50 names). The *aim* $A_{i,t}$ is the centred rank of the prediction over all predicted names, $(R_i - \frac{1}{2})/n - \frac{1}{2}$, or for the multi-horizon blends the mean of the W, M and Q (or W and M) centred ranks, slower horizons carried forward from their latest formation. The aim is re-ranked over m_t to c_i; with σ_i the 63-session realised volatility (missing to the median, floored at a fixed 5th-percentile constant), four maps give the weights:

$$u_i = c_i/\sigma_i, \qquad \mathcal{N}(v)_i = \frac{2\,(v_i - \bar{v})}{\sum_{k:\,v_k \neq 0} |v_k - \bar{v}|} \text{ for } v_i \neq 0, \quad \mathcal{N}(v)_i = 0 \text{ for } v_i = 0, \tag{12}$$

$$w^{(1)} = \mathcal{N}(u), \qquad w_i^{(2)} = w_i^{(1)} - \overline{w^{(1)}}_{g(i)}, \qquad w^{(3)} = \mathcal{N}(w^{(2)}), \tag{13}$$

$$a = \mathcal{N}\big(w^{(3)} - X(X^\top X)^{-1} X^\top w^{(3)}\big), \qquad X = [\mathbf{1},\ \beta], \tag{14}$$

$\bar{v}$ the mean over non-zero entries. The group $g(i)$ is the SIC4 group if it has at least three names, else the SIC2 group among the remaining names if it has at least three, else the SIC division. Equation (14) projects onto the complement of $\text{span}\{\mathbf{1}, \beta\}$, β the 252-session beta to the equal-weight universe, so $\sum_i a_i = 0$, $\sum_i |a_i| = 2$ and $a^\top \beta = 0$ hold exactly; the industry demeaning holds only approximately after the projection, and for $\theta < 1$ the held book mixes targets built on earlier betas, so its beta exposure is small but not zero.

Partial adjustment, cost and statistics. With $\theta \in \{0.05, 0.1, 0.2, 0.5, 1\}$ and $\mathbb{1}_{m_t}$ the tradable mask,

$$w_t = \mathbb{1}_{m_t} \odot \big(w_{t-1} + \theta(a_t - w_{t-1})\big), \qquad g_t = w_t^\top y_t^{raw}, \qquad \tau_t = \|w_t - w_{t-1}\|_1, \qquad \text{net}_t = g_t - c\,\tau_t. \tag{15}$$

Partial adjustment trades a mean information lag of $(1-\theta)/\theta$ periods against turnover, and because $w_0 = 0$ and masking only removes weight, $\sum_i |w_i| \leq 2$ at all times. Turnover is charged against undrifted previous weights; drift-charging changes every computed graph-cell Sharpe ratio by at most 0.01. With $P = 52$,

$$\widehat{\text{SR}} = \frac{\bar{x}}{s_x}\sqrt{P}, \qquad \text{ann} = \bar{x}\,P, \qquad \text{MaxDD} = \min_k \Big(\frac{C_k}{\max_{j \leq k} C_j} - 1\Big), \quad C_k = \prod_{s \leq k}(1 + x_s). \tag{16}$$

Grid, selection and pass rules. The grid has 55 cells, crossing the aims R1 (weekly), R2 (quarterly, carried forward and traded weekly), R4 (W+M+Q) and R4b (W+M) with the universes ALL, LIQ and TOP500 (R2 has no TOP500) and five values of θ. The declared cost per side is 15 bps for ALL, 5 bps for LIQ and 2 bps for TOP500, and every cell is also stored at 1, 2, 5 and 15 bps. *Selection* picks, among cells with development MaxDD $\geq -25\%$ at the declared cost, the one with the highest development net Sharpe ratio. A cell *passes* if its net SR ≥ 1.5 on development *and* on the check window with MaxDD $\geq -25\%$ on both. A graph cell additionally needs, on every horizon its aim uses, a development paired ΔIC > 0 (demeaned basis, Section 7) with month-level $t \geq 2$ against both C0 and C4, and would be relayed only after an independent adversarial review with three pre-agreed reproductions (retraining without price-level features, drift-charged turnover, block-bootstrap standard errors). Both rules were written before the corresponding books were read: for the graph programme before any graph book, and for the 55-cell grid before its read on the rebuilt panels (the same grid had been read on development only on an older build).

7 Inference

Rank information coefficient. Per date, IC_d is the Spearman correlation between prediction and y^{ex} with average ranks on both sides; a date with a constant prediction contributes zero (a stable sort would otherwise rank the constant alphabetically and produce a spurious ± 0.02). Over D dates,

$$\overline{\mathrm{IC}} = \tfrac{1}{D}\sum_d \mathrm{IC}_d, \qquad t = \frac{\overline{\mathrm{IC}}}{s_{\mathrm{IC}}}\sqrt{D}, \qquad \mathrm{ICIR} = \overline{\mathrm{IC}}/s_{\mathrm{IC}}. \tag{17}$$

This t treats dates as independent (the naive t); a cluster-robust version with refit years as clusters is reported where the number of clusters allows.

Paired ΔIC. For cells A, B with seed-mean predictions, $\Delta_d = \mathrm{IC}_d(A) - \mathrm{IC}_d(B)$ on two bases: *raw* (prediction against y^{ex}) and *demeaned* (both demeaned within the SIC4 $\to$ SIC2 $\to$ division cascade of eq. (13), then Spearman). The demeaned basis asks whether the graph adds information *within* industries, which is the only information the industry-neutral book can monetise. Formation values are averaged within calendar months and $t = \bar{\Delta}_{month}/(s_{month}/\sqrt{n_{months}})$.

Sharpe-ratio uncertainty. The iid standard error of an annualised Sharpe ratio from n weekly observations (Lo, 2002) is

$$SE(\widehat{\mathrm{SR}}) = \sqrt{\frac{1 + \widehat{\mathrm{SR}}_w^2/2}{n}}\sqrt{52}, \qquad \widehat{\mathrm{SR}}_w = \widehat{\mathrm{SR}}/\sqrt{52}; \tag{18}$$

a block bootstrap with circular 13-week blocks and 2000 draws gives a dependence-robust alternative (cf. the stationary bootstrap of Politis and Romano 1994; Ledoit and Wolf 2008). Windows are pooled through the exact total-variance identity. For monthly books a block bootstrap of the *centred* series (1000 draws) gives $p = (\#\{\bar{x}^* \geq \bar{x}\} + 1)/(B + 1)$ and a minimum detectable effect $\mathrm{MDE} = P\,(q_{0.95}(\bar{x}^*) - q_{0.20}(\bar{x}^*))$; a within-date permutation of the predictions through the same book code gives a second null, evaluated on the gross statistic because random books turn over more and would otherwise win on cost.

Selection across trials. The expected maximum of N independent null Sharpe ratios with standard error s, $\mathbb{E}[\max_{k \leq N} s\, Z_k]$ (20,000 draws), is the bar a selected best cell has to clear. The deflated Sharpe ratio (Bailey and López de Prado, 2014) is

$$\mathrm{SR}_0 = \sqrt{V}\Big[(1-\gamma)\Phi^{-1}(1-\tfrac{1}{N}) + \gamma\,\Phi^{-1}(1-\tfrac{1}{Ne})\Big], \qquad \mathrm{DSR}_i = \Phi\!\left(\frac{(\widehat{sr}_i - sr_0)\sqrt{n_i - 1}}{\sqrt{1 - \hat{\kappa}_3\widehat{sr}_i + \frac{\hat{\kappa}_4 - 1}{4}\widehat{sr}_i^2}}\right), \tag{19}$$

γ the Euler-Mascheroni constant, V the sample variance of the annualised net Sharpe ratios across the cells that ran, N the declared count, per-period Sharpe ratios inside Φ, and $\hat{\kappa}_3, \hat{\kappa}_4$ the cell's skewness and non-excess kurtosis. The superior-predictive-ability test (White, 2000; Hansen, 2005) is the correctly sized correction when every trial's return series is stored (in our calibration on synthetic series DSR(N) was conservative and SPA well sized). The 349-cell programme stored only per-cell summaries, so the SPA test cannot be applied to it.

From information to Sharpe: cost ladder and IC multiple. For a weekly cell with gross annual return μ_g, gross Sharpe ratio S_g, weekly turnover τ_w and declared cost c_d per side, the stored net Sharpe ratios at 1, 2, 5 and 15 bps are reproduced to within 0.02 by

$$\sigma = \frac{\mu_g}{S_g}, \qquad \mathrm{drag}(c) = 52\,\tau_w\,c, \qquad \mathrm{SR}_{net}(c) = \frac{\mu_g - \mathrm{drag}(c)}{\sigma}, \qquad \hat{c} = \frac{\mu_g}{52\,\tau_w}, \tag{20}$$

$\hat{c}$ the break-even cost. Inverting gives the gross Sharpe ratio needed for a net target $S^\star$ at the declared cost and its ratio to the realised gross Sharpe ratio,

$$S_g^{need} = S^\star + \frac{52\,\tau_w\,c_d}{\sigma}, \qquad \kappa = \frac{S_g^{need}}{S_g}, \tag{21}$$

which we call the IC multiple. At fixed construction, volatility and turnover, gross return scales to first order with the IC (Grinold, 1989; Grinold and Kahn, 2000), so κ is a lower bound on the factor by which the information would have to grow. If $S_g < S^\star$ in every cell, it follows that no cost level reaches the target, since $\mathrm{SR}_{net}(c) \le S_g$ for $c \ge 0$. The same quantities give an ex-post analogue of the transfer coefficient of Clarke et al. (2002), the ratio of the best realised gross Sharpe ratio to the zero-cost ICIR of the raw prediction. Factor exposures of realised books are checked by yearly OLS of the daily spread on [1, Mkt-RF, SMB, HML, MOM] with Newey-West errors (Fama and French, 1993; Carhart, 1997; Newey and West, 1987).

8 Results

Each book in Tables 4 and 9 and each selected book in the text is reported with its net Sharpe ratio, annualised return and maximum drawdown, and every number in a table is generated from the stored result files (Appendix B).

8.1 The information is real, small and replicates

Table 3 gives the fixed booster's rank IC by window and horizon, which is +0.024 weekly ($t = 5.4$) on development and +0.051 ($t = 5.5$) on the check window, +0.023 and +0.050 monthly, and +0.034 and +0.061 quarterly (t per formation). The first read of 2019 to 2022, taken before any portfolio cell existed, gave +0.057 ($t = 3.4$, 47 months) with ridge at +0.048; the single hold-out read of 2023 (eleven months, rule frozen in advance) gave +0.087 ($t = 3.2$) with ridge at +0.057. The sign therefore replicated twice outside the selection data, and because the linear control replicated with it, the replication is a property of the characteristic set rather than of the booster. The signal is ordinal, since out-of-sample R^2 is within ±0.001 of zero on both windows and +0.012 on 2023. Whether fitting adds anything cannot be told on this panel, because composites with no fitted magnitudes (literature signs, textbook five, fitted signs) read IC +0.020, +0.016 and +0.025 over 2009 to 2022 against the booster's +0.032, with paired $t \le 1.4$. With per-refit selection on its validation slice the same model class reads +0.014, no better than ridge (ΔIC = +0.001). In Figure 2 the yearly IC is positive in nine of ten development years (2013 is zero to three decimals) and higher on average on 2019 to 2023, with 2023 just below 2021 (+0.091). The rise is mostly a composition effect, because within the cohort of firms present in 2009 there is no detectable trend after 2014, and the 2019 to 2022 excess sits in post-2009 entrants (cohort +0.023, entrants +0.070), whose survivorship filtering differs.

8.2 No construction turns it into the target Sharpe ratio

Grid. Table 4 lists the best development cells at their declared costs with their check-window values. The pre-declared selection, the W+M blend on the 500 most liquid names at 2 bps, nets 0.63 on development (1.9 % per year, MaxDD −11.9 %, gross 1.07) and 0.28 on the check window (1.7 %, −8.8 %, gross 0.49). The pass bar of 1.5 is missed, and no cell even reaches 0.5 on both windows (0 of 55). Figure 3 places every cell by gross and net Sharpe ratio. The cloud sits below 1.5 on the gross axis in both windows, and the best development cell is not the best check cell (the cross-window rank correlation of the 55 cells is 0.26 net and 0.55 gross), which is what selection on a noisy but not empty ordering produces. By the iid standard errors of Table 5, 0.32 and 0.51, the target is 2.71 and 2.39 standard errors away; pooled over 2009 to 2022 the selected cell nets 0.45 ± 0.27 with upper 95% limit 0.98, and none of the three net intervals excludes zero.

Table 3: Rank IC of the fixed booster by window and horizon, with the linear control where run. $t = \overline{\text{IC}}/s_{\text{IC}}\sqrt{n}$ over the n formation periods, treating periods as independent.

Window	Horizon	Model	Mean rank IC	t	n periods
dev 2009-2018	weekly	fixed booster (grid panels)	+0.024	5.4	503
check 2019-2022	weekly	fixed booster (grid panels)	+0.051	5.5	201
dev 2009-2018	monthly	fixed booster (grid panels)	+0.023	2.7	120
check 2019-2022	monthly	fixed booster (grid panels)	+0.050	3.1	47
dev 2009-2018	quarterly	fixed booster (grid panels)	+0.034	2.3	40
check 2019-2022	quarterly	fixed booster (grid panels)	+0.061	2.1	15
first read 2019-2022	monthly	fixed booster (first read)	+0.057	3.4	47
first read 2019-2022	monthly	ridge-all control (first read)	+0.048	2.9	47
hold-out 2023	monthly	booster (hold-out)	+0.087	3.2	11
hold-out 2023	monthly	ridge-all control (hold-out)	+0.057	2.3	11

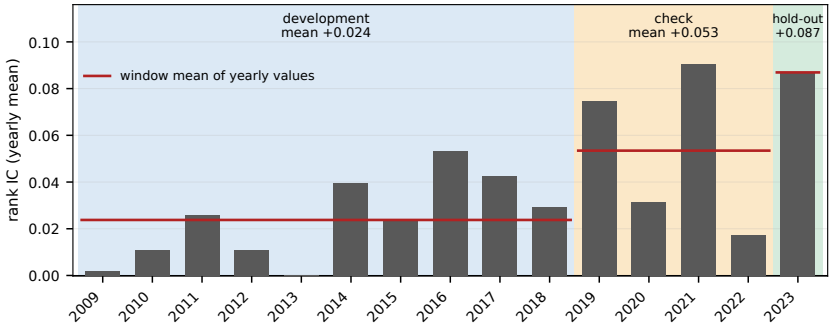


Figure 2: Yearly mean of the monthly rank IC of the fixed booster used for the out-of-window and hold-out reads, 2009 to 2023; development, check and hold-out windows shaded, with the mean of each window's yearly values.

Table 4: Best development cells of the 55-cell grid at the declared cost of their universe (ALL 15, LIQ 5, TOP500 2 bps per side), development MaxDD $\geq -25\%$. † marks the pre-declared selection. Last row: counts over all 55 cells.

	Development 2009-2018				Check 2019-2022				
Cell	gross SR	net SR	ann. %	MaxDD %	gross SR	net SR	ann. %	MaxDD %	turnover/wk (dev)
R4b W+M / TOP500 / θ=1.0†	1.07	0.63	+1.9	−11.9	0.49	0.28	+1.7	−8.8	1.298
R4 W+M+Q / TOP500 / θ=1.0	0.95	0.61	+1.9	−10.4	0.41	0.25	+1.6	−11.8	1.030
R4b W+M / TOP500 / θ=0.5	0.82	0.58	+1.5	−10.6	0.51	0.40	+2.1	−10.4	0.613
R1 weekly / TOP500 / θ=0.5	0.89	0.55	+1.4	−10.0	0.53	0.38	+2.0	−6.1	0.823
R4 W+M+Q / TOP500 / θ=0.5	0.67	0.50	+1.5	−9.6	0.39	0.30	+1.8	−12.9	0.500
R1 weekly / TOP500 / θ=1.0	1.09	0.48	+1.6	−11.3	0.58	0.31	+2.0	−6.3	1.895
R4 W+M+Q / ALL / θ=0.05	0.72	0.42	+0.8	−3.9	0.73	0.62	+3.2	−13.6	0.074
R1 weekly / TOP500 / θ=0.2	0.53	0.36	+0.8	−8.9	0.45	0.38	+1.7	−8.2	0.331

Of 55 cells at the declared cost: net SR ≥ 0.5 on both windows: 0; net SR ≥ 1.5 on both windows: 0; max gross SR dev 1.37 (R1 weekly / ALL / θ=1.0), chk 0.97 (R1 weekly / ALL / θ=1.0). † = selected cell.

Selection bar and transfer. The expected maximum of N independent zero-mean null Sharpe ratios over ten years is 0.91 at $N = 294$ (stored) and 0.93 at $N = 349$ (simulated), and the best development net Sharpe ratio on the rebuilt panels, 0.63 (0.83 on the superseded older build), is below it. The net books alone therefore cannot be told from selection noise, while the gross books (selected 1.07, best 1.37) and the annualised ICIR of the weekly prediction, 1.72 ($t = 5.4$), are above it, which puts the distance between the information and the net book partly in transfer and partly in cost. The bar is only indicative, because the cells are correlated, which lowers it, and because a no-information net Sharpe ratio has mean $-\text{drag}/\sigma$. The annualised ICIR of the raw weekly prediction ($t\sqrt{52/n}$) is 1.72 on development and 2.78 on the check window; the highest gross Sharpe ratio any of the 55 constructions realises is 1.37 and 0.97, a measured transfer coefficient of 0.80 and 0.35. Both are upper estimates (the numerator is a maximum over 55 cells), the check ratio carries ±0.18 from the Sharpe standard error alone, and the book is volatility-scaled and neutralised over the tradable set while the ICIR scores every name, so the ratio is an ex-post analogue of a transfer coefficient. Gross Sharpe ratios rise with trading and net ones fall with it, as the weekly-traded $\theta = 1$ books (R1, R4, R4b) show.

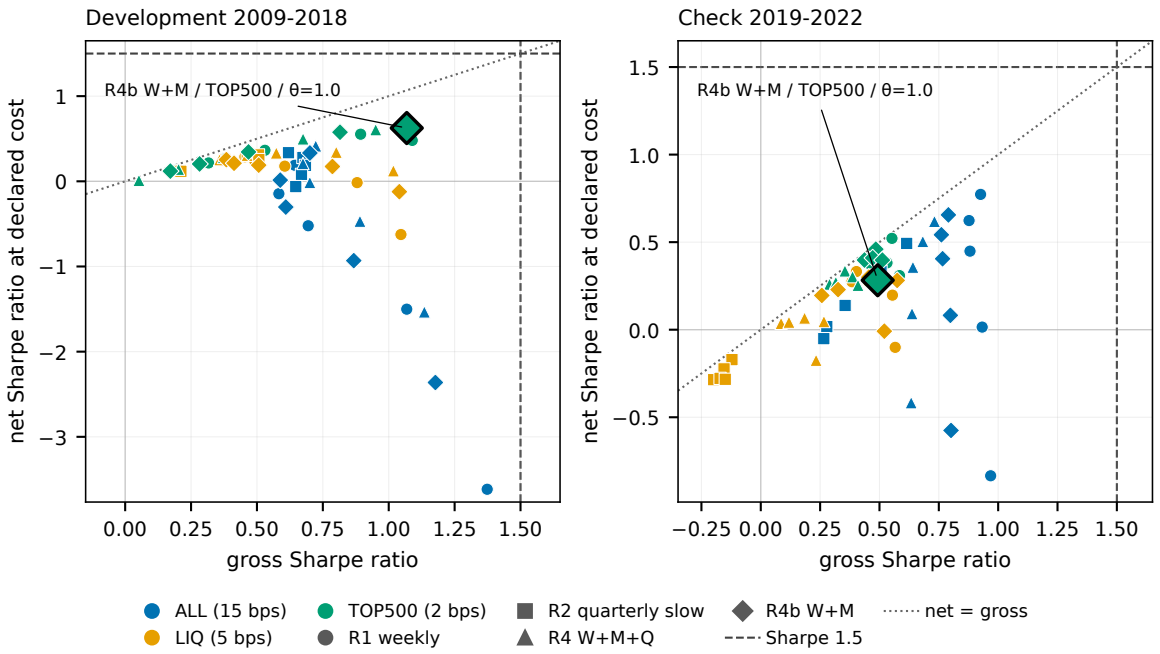


Figure 3: Gross against net Sharpe ratio at the declared cost for the 55 cells, development (left) and check window (right); colour by universe, marker by aim family, the selected cell outlined; dashed lines at 1.5, dotted line net = gross.

Table 5: Precision of the selected cell's Sharpe ratios: iid standard error (eq. 18), 95% interval, distance to the target 1.5 and to zero in standard errors; counts over the 55 cells of pooled upper limits reaching 1.0 and 1.5.

Window		SR	weeks	iid SE	95% CI	z to 1.5	t to 0
dev	net	0.63	503	0.32	[−0.01, 1.26]	2.71	1.94
dev	gross	1.07	503	0.32	[0.43, 1.70]	1.34	3.30
chk	net	0.28	201	0.51	[−0.71, 1.28]	2.39	0.56
chk	gross	0.49	201	0.51	[−0.51, 1.49]	1.98	0.97
pooled	net	0.45	704	0.27	[−0.08, 0.98]	3.86	1.66
pooled	gross	0.77	704	0.27	[0.24, 1.31]	2.66	2.84

Of 55 cells, pooled net upper 95% bound ≥ 1.5: 0 (none); ≥ 1.0: 1 (R4b W+M / TOP500 / θ=0.5); pooled gross upper bound ≥ 1.5: 1 (R1 weekly / ALL / θ=1.0).
Maximum pooled net SR: 0.47 (R4b W+M / TOP500 / θ=0.5), 95% CI [−0.06, 1.00]. SE is iid (Lo 2002); the selected cell is the maximum over 55 cells.

They reach the highest gross Sharpe ratios at one to two turns a week, and on development they are net-positive at 1 to 2 bps, between −0.62 and +0.24 at 5 bps and deeply negative at 15 bps. The slow ALL books of the weekly-traded aims (R1, R4 and R4b at $\theta = 0.05$) keep about half to two thirds of the development gross Sharpe ratio at under a tenth of the turnover (Table 4, Figure 3).

Crisis years in the check window. The check window contains the March 2020 crash and the 2022 bear market. Table 6 splits it by calendar year for three series whose yearly values are stored: the monthly rank IC, the monthly decile book of Section 9, and a full-breadth rank-weighted book made dollar-, beta- and SIC-division-neutral and held in three overlapping monthly cohorts (both ALL, 15 bps); the weekly grid books are stored only as window summaries. The 2020 yearly IC, +0.031, is close to the development mean +0.024 and has a standard error of 0.032, so the year neither confirms nor excludes a failure. The crash months are negative (February −0.160, the 4th lowest of 168 months, and March −0.079) and April recovers (+0.239). The decile book returned −5.6 % in February and −7.3 % in March, +7.6 % in April, and ended 2020 at +4.7 %. The weakest check year for the IC is 2022 (+0.017); without 2020 the check mean of yearly ICs moves from +0.053 to +0.061. The decile book's worst year is 2009 (−23.7 %), inside the development window; the neutral book's worst year is 2020 (−13.3 %, against +13.5 % in 2022), so the size and sign of a crisis-year loss depend on the construction. The check window was the better window for the information and for the decile book's

Table 6: The check window by calendar year (post-hoc split): yearly mean of the monthly rank IC of the fixed booster, and calendar-year net return of the monthly decile long-short book (ALL, 15 bps) and of a neutral rank-weighted book held in three-month cohorts.

Year	rank IC	decile book net %/yr	neutral cohort book net %/yr
2019	+0.075	+14.4	+3.6
2020	+0.031	+4.7	−13.3
2021	+0.091	+15.8	+15.5
2022	+0.017	−1.2	+13.5
mean, development 2009-2018	+0.024	−0.2	−0.3
mean, check 2019-2022	+0.053	+8.4	+4.8
mean, check without 2020	+0.061	+9.7	+10.9
worst decile-book year (2009)	+0.002	−23.7	−9.7

Yearly mean of the monthly rank IC of the fixed booster; calendar-year net return of the monthly decile long-short book (ALL, 15 bps per side) and of a full-breadth rank-weighted book made dollar-, beta- and SIC-division-neutral and held in three overlapping monthly cohorts (ALL, 15 bps). Post-hoc split of a pre-declared window; means are of yearly values.

Table 7: Cost-ladder inversion (eq. 21) for the selected cell and the least demanding cells: annual volatility σ, turns per year, declared cost c_d and break-even cost $\hat{c}$ per side, the gross Sharpe ratio S_g^{need} needed for net 1.5 at c_d, and the information multiple κ.

Case, window	Cell	gross SR	σ %	turns/yr	c_d bps	$\hat{c}$ bps	S_g^{need}	κ
selected, dev	R4b W+M / TOP500 / θ=1.0	1.07	3.0	67.5	2	4.8	1.94	1.82
selected, chk	R4b W+M / TOP500 / θ=1.0	0.49	5.9	62.2	2	4.7	1.71	3.47
least demanding (dev), dev	R4b W+M / TOP500 / θ=1.0	1.07	3.0	67.5	2	4.8	1.94	1.82
least demanding (chk), chk	R1 weekly / ALL / θ=0.05	0.93	4.4	4.5	15	90.7	1.65	1.79
least demanding (both), dev	R4 W+M+Q / ALL / θ=0.05	0.72	1.9	3.8	15	35.3	1.81	2.50
least demanding (both), chk	R4 W+M+Q / ALL / θ=0.05	0.73	5.1	3.8	15	97.3	1.61	2.20

Cells where a positive cost would give net 1.5: dev 0, chk 0. Max ladder reproduction error: 0.0186 Sharpe. σ = annualised book volatility; c_d declared, $\hat{c}$ break-even cost per side; S_g^{need} = gross SR needed for net 1.5 at c_d; κ = first-order IC multiple needed.

return (+8.4 % per year against −0.2 % on development) but not for its risk, since volatility rises from 11.2 % to 22.3 %. The weekly grid summaries also point to a higher-volatility rather than a lower-return window: for the selected cell gross return is 3.3 % per year on development and 2.9 % on the check window while volatility rises from 3.0 % to 5.9 %; across the 55 cells the check-to-development volatility ratio is 1.7 to 2.9 (median 2.1), gross return is higher on the check window in 41 cells, and 40 cells have a deeper check-window drawdown at the declared cost despite the shorter window. This is consistent with a crisis effect on the weekly Sharpe ratios but does not establish one. The ex-2020 figures are a post-hoc diagnostic and the pass rule is unchanged. A predictor restricted to ordinary conditions would need an ex-ante trigger, say on volatility or drawdown, and testing it would be a new trial that had to be declared in advance and was not.

8.3 What a net Sharpe ratio of 1.5 would take

Equation (20) reproduces the stored cost ladder of all 55 cells to within 0.02, so its inversion, eq. (21), is arithmetic on the stored summaries (Table 7). No cell's gross Sharpe ratio reaches 1.5 on either window (maximum 1.37 development, 0.97 check), so no cost level, including zero, closes the gap; the cost that would is negative for every cell. The information multiple is 1.82 (development) and 3.47 (check) for the selected cell and 2.50 for the least demanding cell on both windows, so the gap is one of information, and no lowering of costs can close it. The multiple is only a lower bound, and a larger IC need not buy a larger Sharpe ratio, because between the two windows the least demanding cell's weekly IC doubled (+0.024 to +0.051) while its gross Sharpe ratio stayed at 0.72 and 0.73. The selected cell's break-even cost on development is 4.8 bps per side; at its declared 2 bps the drag takes about four tenths of the gross.

Table 8: Graph programme: seed-mean rank IC by cell and horizon, development (t) / check (t); naive per-formation t. "pending" marks cells not run before the pause.

Cell	W	M	Q
C0	+0.018 (+3.7) / +0.052 (+5.2)	+0.017 (+1.7) / +0.054 (+3.7)	+0.005 (+0.3) / +0.071 (+3.0)
C0g	+0.017 (+3.1) / +0.053 (+4.9)	+0.016 (+1.6) / +0.056 (+3.2)	+0.016 (+0.9) / +0.038 (+1.1)
C1	+0.017 (+3.5) / +0.052 (+5.0)	+0.018 (+1.9) / +0.057 (+3.9)	+0.012 (+0.7) / +0.051 (+2.0)
C2	+0.014 (+2.4) / +0.053 (+4.8)	+0.014 (+1.2) / +0.055 (+2.8)	+0.019 (+1.0) / +0.035 (+0.9)
C3	+0.013 (+2.2) / +0.053 (+4.7)	+0.013 (+1.0) / +0.059 (+3.0)	+0.012 (+0.6) / +0.044 (+1.1)
C4	pending	+0.021 (+2.2) / +0.055 (+3.5)	+0.016 (+0.8) / +0.047 (+1.6)
C5	pending	+0.018 (+1.6) / +0.053 (+2.7)	0.000 (0.0) / +0.056 (+1.4)

Cell values: dev 2009-2018 IC (t) / check 2019-2022 IC (t); seed means; naive per-formation t.

Table 9: Graph programme: the selected book of each prediction set under the 55-cell grid and rules of Section 6; net Sharpe ratio (annualised return, MaxDD), gross Sharpe ratio, standard errors (iid / 13-week block bootstrap). Eligible cells are those the rule may select.

Cell	eligible	selected book	dev net	chk net	dev gross	chk gross	SE dev	SE chk
C0	no	R1 weekly / TOP500 / θ=0.5	0.55 (+1.7, −13.6)	0.18 (+1.1, −9.8)	0.82	0.31	0.32 / 0.46	0.51 / 0.37
C0g	no	R1 weekly / TOP500 / θ=0.5	0.41 (+1.3, −11.8)	−0.21 (−1.4, −15.4)	0.67	−0.10	0.32 / 0.42	0.51 / 0.37
C1	yes	R4b W+M / TOP500 / θ=1.0	0.52 (+1.7, −12.8)	−0.06 (−0.4, −11.4)	0.90	0.14	0.32 / 0.41	0.51 / 0.43
C2	yes	R1 weekly / TOP500 / θ=0.5	0.25 (+0.8, −11.0)	0.08 (+0.5, −8.8)	0.49	0.21	0.32 / 0.40	0.51 / 0.38
C3	yes	R1 weekly / TOP500 / θ=0.5	0.02 (+0.1, −16.9)	0.25 (+1.5, −8.1)	0.26	0.37	0.32 / 0.40	0.51 / 0.37
C4	no	pending	pending	pending	pending	pending	pending	pending
C5	yes	pending	pending	pending	pending	pending	pending	pending
booster (ref.)	n/a	R4b W+M / TOP500 / θ=1.0	0.63 (+1.9, −11.9)	0.28 (+1.7, −8.8)	1.07	0.49	n/a	n/a

Net SR (annualised return %, MaxDD %) at the declared cost of the book's universe; SE = Sharpe standard error, iid / 13-week block bootstrap; bold = best eligible cell.

8.4 The completed graph networks add no detectable information

Levels. Table 8 gives the seed-mean rank IC of each graph cell by horizon and window, Table 9 the selected book of each prediction set under the identical grid and rules. The no-graph baseline C0 nets 0.55 on development and 0.18 on the check window, against 0.52 and −0.06 for C1, 0.25 and 0.08 for C2, 0.02 and 0.25 for C3, and 0.63 and 0.28 for the booster on the same grid. On development every graph book is at or below the baseline (on the check window C3 is above it, 0.25 against 0.18, inside one standard error of 0.51); all are far from 1.5.

Paired ΔIC. Table 10 and Figure 4 carry the primary evidence. On development the demeaned ΔIC of C1 against C0 is small and of mixed sign across horizons, and C2 and C3 are slightly negative on the weekly horizon. The largest $|t|$ against C0 on development is 1.9, below the pre-declared threshold of 2. Only one comparison has $t \geq 2$, C1 against the aggregate-feature control C0g on the weekly horizon, and it is not a pass condition and does not repeat on the check window. The only other $|t| \geq 2$ is negative (C1 against C0 on the quarterly check window, raw basis).

Rewired null and label shift. C4, where it ran, has the highest monthly development IC of any graph cell and quarterly and check-window ICs inside the range of the real-graph cells, which is what one expects if message passing fits degree and industry composition rather than specific pairings; its paired comparison was not completed. None of the four label-shift diagnostics meets the persistence rule, but at these ICs the diagnostic has little power (the real development ICs have $t \leq 1.7$, and on the quarterly horizon the shifted models read higher ICs than the real ones), so it neither attributes nor clears a signal that is not there.

Scope. The following were not completed before the pause: the weekly C4 and C5 predictions, the C4 and C5 books, every paired ΔIC against C4, all of C5's paired ΔIC, the weekly label-shift runs and the formal selection script. Because a pass needs a development ΔIC against C0 with $t \geq 2$ on every horizon used, and no horizon of C1 to C3 has one, the missing C4 comparisons cannot make C1 to C3 pass. C5 has neither a book nor a paired ΔIC; its monthly and quarterly ICs sit at the baseline, so a pass is unlikely but not excluded. For the completed cells the stop rule's negative form applies, so C1 to C3, trained end to end, add a ΔIC that cannot be told from zero over the same features without a graph, and the gap to the target remains the information gap of Section 8.3.

Table 10: Paired ΔIC of each graph cell against the controls (C0 no-graph; C0g aggregate features), by horizon and window, demeaned and raw bases, month-level t in parentheses; $*$ marks development demeaned $|t| \geq 2$.

Cell	control	h	dev demeaned	chk demeaned	dev raw	chk raw
C1	C0	W	+0.0015 (t +1.0)	−0.0023 (t −1.1)	−0.0013 (t −0.7)	−0.0006 (t −0.2)
C1	C0	M	+0.0041 (t +1.4)	+0.0079 (t +1.8)	+0.0010 (t +0.2)	+0.0030 (t +0.5)
C1	C0	Q	−0.0046 (t −1.0)	+0.0050 (t +0.7)	+0.0068 (t +1.1)	−0.0201 (t −2.2)
C1	C0g	W	+0.0029 (t +2.1) *	−0.0014 (t −1.0)	+0.0003 (t +0.2)	−0.0010 (t −0.5)
C1	C0g	M	+0.0023 (t +0.6)	−0.0017 (t −0.2)	+0.0014 (t +0.4)	+0.0008 (t +0.1)
C1	C0g	Q	−0.0084 (t −1.4)	+0.0149 (t +0.8)	−0.0038 (t −0.5)	+0.0126 (t +0.7)
C2	C0	W	−0.0029 (t −1.9)	−0.0015 (t −0.3)	−0.0039 (t −1.5)	+0.0010 (t +0.3)
C2	C0	M	+0.0046 (t +0.9)	+0.0027 (t +0.3)	−0.0032 (t −0.5)	+0.0007 (t +0.1)
C2	C0	Q	+0.0005 (t +0.1)	−0.0033 (t −0.1)	+0.0138 (t +1.4)	−0.0363 (t −1.2)
C2	C0g	W	−0.0015 (t −1.5)	−0.0006 (t −0.1)	−0.0022 (t −1.2)	+0.0005 (t +0.3)
C2	C0g	M	+0.0028 (t +0.9)	−0.0068 (t −1.7)	−0.0028 (t −0.7)	−0.0015 (t −0.3)
C2	C0g	Q	−0.0034 (t −0.8)	+0.0066 (t +0.8)	+0.0032 (t +0.6)	−0.0037 (t −0.5)
C3	C0	W	−0.0037 (t −1.7)	−0.0023 (t −0.4)	−0.0050 (t −1.3)	+0.0005 (t +0.1)
C3	C0	M	+0.0012 (t +0.3)	+0.0062 (t +0.7)	−0.0038 (t −0.5)	+0.0049 (t +0.4)
C3	C0	Q	−0.0023 (t −0.3)	+0.0033 (t +0.1)	+0.0067 (t +0.6)	−0.0263 (t −0.9)
C3	C0g	W	−0.0022 (t −1.5)	−0.0014 (t −0.3)	−0.0034 (t −1.2)	0.0000 (t +0.1)
C3	C0g	M	−0.0005 (t −0.1)	−0.0034 (t −0.7)	−0.0033 (t −0.6)	+0.0027 (t +0.4)
C3	C0g	Q	−0.0062 (t −0.9)	+0.0132 (t +1.3)	−0.0039 (t −0.4)	+0.0064 (t +0.5)

Paired ΔIC (cell minus control) per formation, month-level t. $*$ = $|t| \geq 2$ on the development demeaned basis. Rows with pending C4/C5 results omitted.

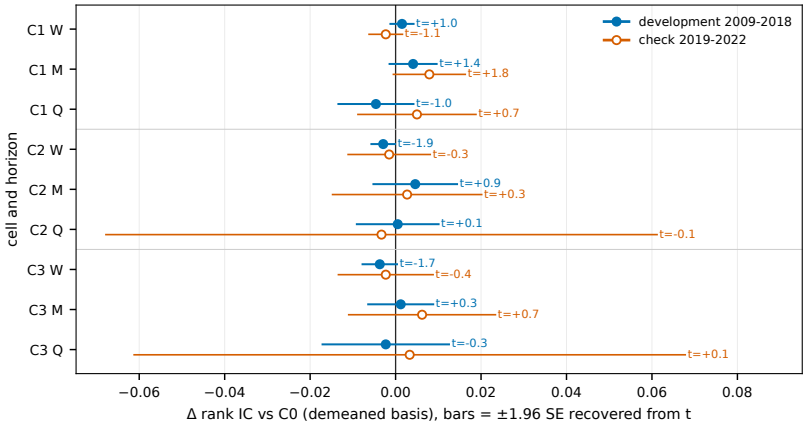


Figure 4: Development paired ΔIC (demeaned basis) of the graph cells against C0 by horizon, with ±1.96 standard errors recovered from the t-statistic (approximate: t is stored to one decimal); hollow markers are the check window.

Table 11: Label-shift diagnostic (seed 0): IC of the model trained on labels shifted back one year against the real model; the declared persistence rule (shifted development IC > 0, $t \geq 2$, at least half the real IC) is not met in any case.

Cell	h	shifted dev	shifted chk	real dev	real chk	shifted/real (dev)	persists
C2	M (K=13)	+0.009 (t +0.9)	+0.001 (t 0.0)	+0.019 (t +1.7)	+0.050 (t +2.5)	0.48	no
C2	Q (K=5)	+0.019 (t +1.0)	−0.012 (t −0.3)	+0.014 (t +0.8)	+0.031 (t +0.9)	1.37	no
C3	M (K=13)	+0.009 (t +0.8)	+0.013 (t +0.6)	+0.012 (t +0.9)	+0.046 (t +2.3)	0.74	no
C3	Q (K=5)	+0.024 (t +1.2)	−0.028 (t −0.7)	+0.007 (t +0.4)	+0.031 (t +0.9)	3.28	no

Persists = shifted dev IC > 0 with $t \geq 2$ and $\geq 0.5\times$ the real dev IC (seed 0).

9 Power and the sizing decision

How much forward data would it take to confirm a signal this small at its own level? We answer from the model's monthly IC series over 2009 to 2022 (168 months; mean +0.032, standard deviation 0.095, lag-one autocorrelation −0.08) and from the Sharpe-ratio standard error of eq. (18). For a one-sided 5% test of H_0: mean IC ≤ 0, 80% power needs 63 months at a true IC of 0.03 and 16 at 0.06. Telling the 2019 to 2023 level (0.06) from the long-run level needs at least as many, and six months of forward data have power 11.3 % and 37.5 % at the two levels. A block bootstrap of the same calculation agrees

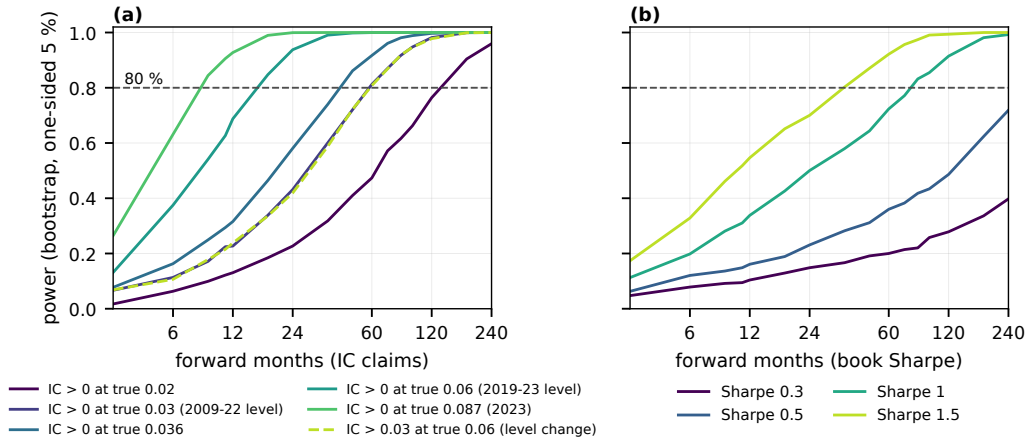


Figure 5: Power against forward months (log scale) for the IC claims (left) and for Sharpe-ratio levels (right); dashed line at 80%. Block-3 bootstrap from the demeaned real series: the model's 2009 to 2022 monthly IC series (left) and its 2009 to 2023 monthly decile long-short net series at 15 bps (right).

(Figure 5). For a book, confirming a Sharpe ratio of 0.3 against zero takes about 68.8 years of monthly data, a Sharpe ratio of 1.0 about 6.2 years.

The signal is present but cannot be confirmed at its own Sharpe level on any horizon that matters for an allocation decision. One strong year such as 2023 moves the posterior mean little against fourteen years at a Sharpe ratio of 0.1 to 0.2 (monthly decile book 0.18, 2.8 % per year, MaxDD −46.6 %, 2009 to 2022). Sizing such a signal is therefore a prior decision rather than an estimate the data can deliver.

10 Limitations

Survivorship. The price feed lists firms as of October 2026; failures, acquisitions and firms that went dark are absent, which biases a long-short spread in a direction that is not fixed (Shumway, 1997). For the monthly decile legs of the booster (2014 to 2023), on SEC filing dates and 8-K items with fixed return-at-death assumptions (failure −40%, acquisition +20%, dark −20%) and blank-check companies excluded, the central estimate of the final-year channel's understatement of the spread is 0.2 to 0.45 percentage points per year, at most about 2.2 across variants, and its sign is not identified (it flips if the short leg is acquisition-heavy within its bins); the weekly books were not measured, and the level channel (about two thirds of 2013 filers had stopped filing by 2026) is not bounded. The models load on the characteristics most flattered by survivorship (short-term reversal, idiosyncratic volatility); an ablation without them improves the ensemble out of sample, which we report as a diagnostic only. A delisting-inclusive panel is the one change that would settle this.

Reused check window. The 2019 to 2022 window had been read three times before the graph experiments; the graph programme is its fourth use. The 2023 read is the only genuinely held-out year and is eleven months long; the frozen rule has 8 to 12% power at the development IC of 0.024 (14 to 17% at 0.03), so its pass is weak evidence as an event, and the observed IC of +0.087 is the stronger evidence.

Price-level measurement error. The feed's closes are split-adjusted at download date and the panel builder undoes only the splits inside the loaded window, so the $5 filter and the five price-level characteristics (`log_price`, `log_mcap`, `bm`, `ep`, `sp`) are wrong by the product of later splits for about 4 to 7% of firm-months. This is a measurement error tied to future corporate actions rather than look-ahead on returns, and a variant without the filter and the five characteristics replicates the out-of-window IC (+0.050 against +0.057).

Incomplete graph programme. Section 8.4 lists the runs not completed before the pause. The conclusion rests on C0, C0g, C1, C2 and C3 with ICs, books and paired ΔIC against C0 and C0g on all horizons, and on C4 and C5 monthly and quarterly ICs; for C1 to C3 the missing comparisons cannot produce a pass, and C5 remains open.

Library defaults. The booster's λ and γ and the pairwise pair sampling are xgboost 2.1.4 defaults, and the deep models ran under torch 2.14.1. We state these defaults and versions but did not control them.

11 Conclusion

On a free, point-in-time, survivor-only US equity panel, a pre-fixed shallow booster carries a rank signal that replicates out of its development window and on one held-out year, but that signal is too small to become a net Sharpe ratio of 1.5 under any declared construction or cost. The shortfall lies in the information, of which the least demanding cell would need, to first order, at least about 2.50× as much. On the development window the three completed graph networks, trained end to end, add no detectable information over the same features without a graph by paired ΔIC, and the rewired-graph comparison, completed only as monthly and quarterly IC levels, points the same way, while the temporal network C5 has no paired comparison and remains open. What the study leaves standing is its method, namely a validation geometry in which no two labels on opposite sides of a fit, validation or test boundary share a daily return, a cost-ladder identity that separates information from cost, a rewired-graph null, and a power calculation that puts confirmation of a Sharpe-0.3 property decades away. A delisting-inclusive panel and a forward prediction log would change what can be said next.

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A Pitfalls that changed numbers

Each item altered a reported statistic before it was found, and its fix is now part of the protocol.

1. **Constant booster and ties.** Tiny pairwise hessians with `min_child_weight` 500 forbade every split; a stable sort ranked the constant alphabetically (IC ±0.02). The fixes are `min_child_weight` 1, a non-degeneracy assertion, a tie-safe average-rank IC and seeded tie-breaks.
2. **Validation-IC noise.** Standard error 0.024 on a 12-month slice against grid margins of 0.000 to 0.005; early stopping halted at about five trees. The fix is a fixed number of rounds with no per-refit selection.
3. **Rank target not market-neutral.** Correlation 0.75 with Mkt-RF; a validation year in a falling market selects a low-beta book. The fix is an ex-ante beta per year, eq. (14).
4. **Cost-confounded permutation null.** Random decile books turn over 1.6 times more, so beating them on net Sharpe ratio measured lower turnover (net $p = 0.002$, gross $p = 0.42$). The fix is to evaluate the null on the gross statistic.
5. **Survivorship-loaded features.** Reversal and idiosyncratic volatility dominate. The fix is an ablation diagnostic, while a delisting-inclusive panel remains open.
6. **Price levels wrong by future splits.** The $5 filter and the five price-level characteristics are wrong by later splits (Section 10), and the fix is a variant without them.
7. **Forward-looking event flags.** Two earnings-calendar flags used later 8-K filings. The fix is a clean point-in-time event mask.
8. **Cache key and annualisation.** A cache keyed without the split returned another split's predictions; weekly series annualised with $\sqrt{12}$ mis-ranked cells in a deflated Sharpe ratio. The fix puts split and horizon in the key and uses per-horizon periods per year (50.4 / 12 / 4; books use 52, rescaled by $\sqrt{50.4/52}$ in selection).
9. **Validation-selected "no signal".** Six model classes read rank IC at most +0.017 only when validation-selected and early-stopped, so the selector rather than the models caused the result.

B Reproducibility

Every table and figure is generated by two scripts in the accompanying code repository (`paper/build_tables.py`, `paper/build_figures.py`) from stored result files; no table number is typed by hand, and the numbers quoted in the text are macros that the same script writes. The stored sources are the 55-cell grid and its cost ladder (`p12c_eval.json`), the cost-ladder inversion (`p12c_what_it_takes.json`), Sharpe standard errors (`p12c_precision.json`), the expected maximum of N and the zero-cost benchmarks (`p12_stage4.json`), the out-of-window and hold-out IC reads (`p9_e6.json`, `p9_b_wfa.json`, `p9_b_fullhistory.json`), the monthly series (`p9_b_followups.json`) and the power calculation (`p9_b_forward_power.json`), all under `results/`, and the graph programme's generated results file `results/gnn/RESULTS.md` with `STATE.json`. A self-test reproduces the stored cost ladder from eq. (20) for all 55 cells and the stored expected maximum of 294 null Sharpe ratios by simulation.

The graph programme's hyperparameters are one set, hashed (`872821b168bcbcc3`) and registered before any graph result was read. Prediction files carry the panel fingerprint, hyperparameter hash and code hash and must reproduce the exact row sets of the grid (1,347,743 / 318,475 / 103,945 rows), and 808 refits were trained. The pre-registration documents of both programmes are in `docs/`.

Zone attestation. Only training-zone rows were used (formation months 2009-01 to 2022-12, maximum formation date 2022-12-20 weekly). The 2023 walk-forward rows were scored once under a frozen rule (an earlier script loaded and masked them without scoring), and no price or return from the sealed zone from 2024-07-01 has been read (the survivorship diagnostic's 2026-vintage filing-status table reaches into it for two 2023 formations, with no returns). The check window 2019 to 2022 was used as a check four times, counted in Section 3.